



State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

Python Closes the Gap: 8 Years of Language Usage Shifts

The top-5 landscape is actively reshuffling: Python and TypeScript rise while JavaScript and PHP decline.

Key Language Trajectories (2017 - 2024)

JavaScript: Declining from Peak

Peaked at 70% in 2020, declining steadily to 61% in 2024.
Still the most used overall, but losing absolute dominance.

Python: Steady Strategic Growth

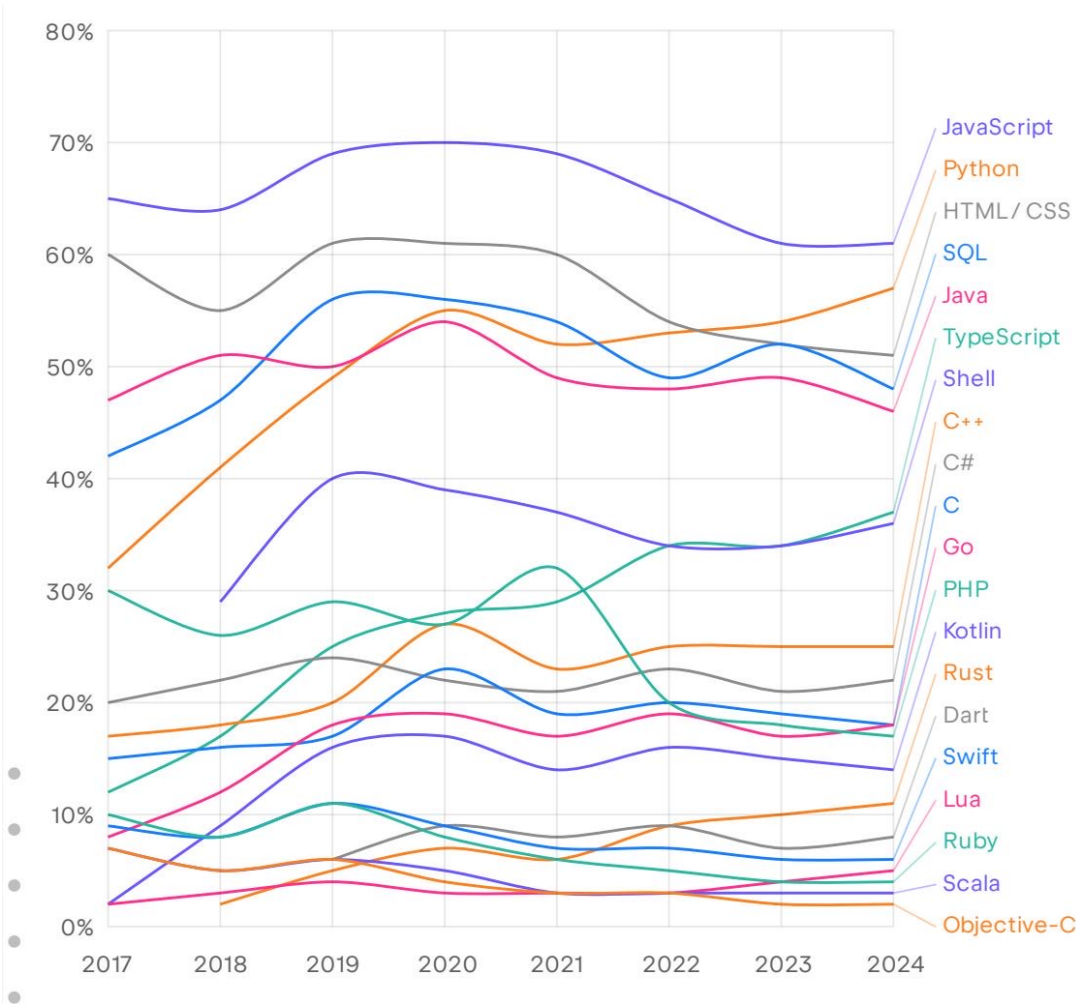
Rose steadily from 32% (2017) to 57% (2024).
Driven by AI/ML boom; nearly closing the gap with JS.

TypeScript: Tripling its Share

Massive growth from 12% (2017) to 37% (2024).
Reflects enterprise demand for safe frontend development.

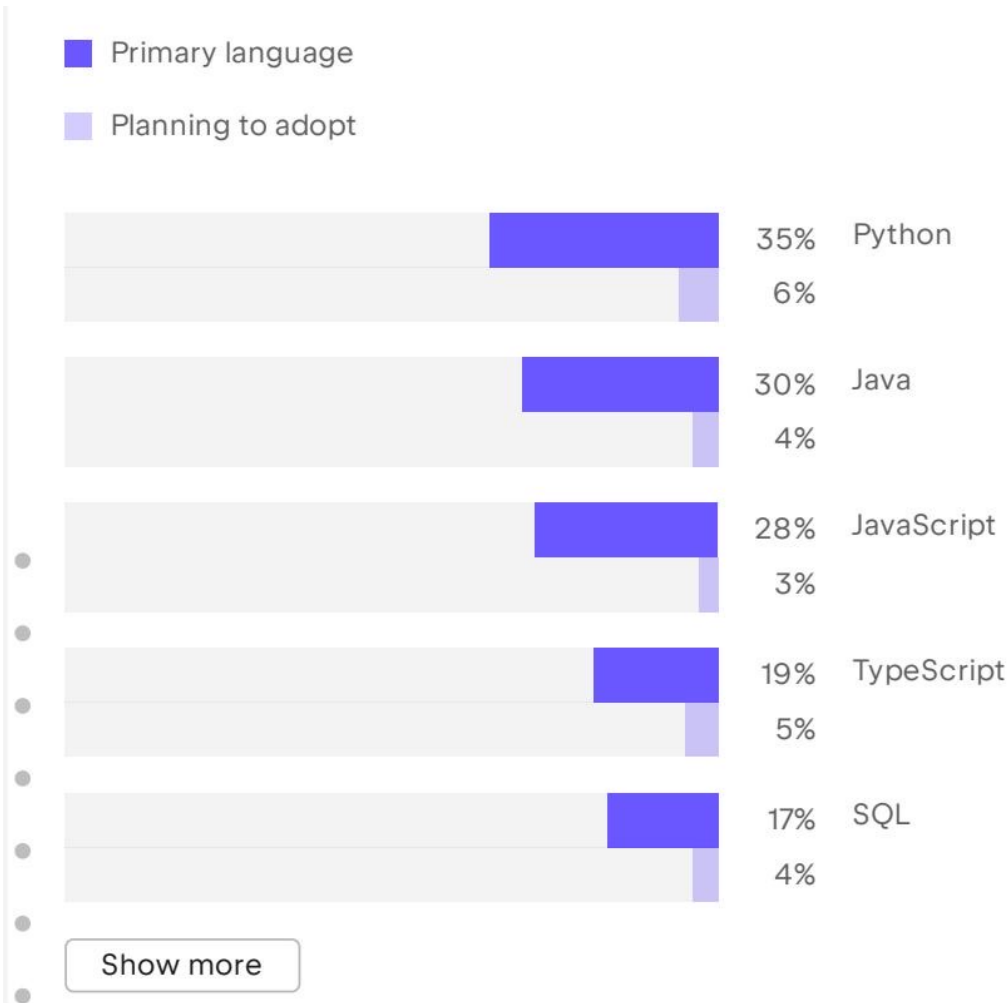
PHP: Significant Drop

Dropped significantly from 30% (2017) to 17% (2024).
Losing ground to newer web framework alternatives.



Python Is Now the #1 Primary Language at 35%

Ecosystem lock-in effect: Despite shifting overall usage, adoption intent for new primary languages remains below 6%.



Primary Usage Breakdown (2024)

1. Python (35%)

Leads as the primary language for 35% of developers.

2. Java (30%)

Maintains a strong hold as the primary enterprise backend stack.

3. JavaScript (28%) & TypeScript (19%)

Combined web frontend ecosystem captures nearly half the market.

The "Comfort Zone" Effect

Adoption Intent Below 6%

Only 6% plan to adopt Python, 5% TypeScript, and 4% Java/SQL.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Type-safe, modern-tooling languages dominate the top spots, reflecting shifting developer expectations.

Key Satisfaction Drivers

1. TypeScript (199 Score)

Brings safety and tooling to JavaScript.
The undisputed leader in developer love.

2. Rust (198 Score)

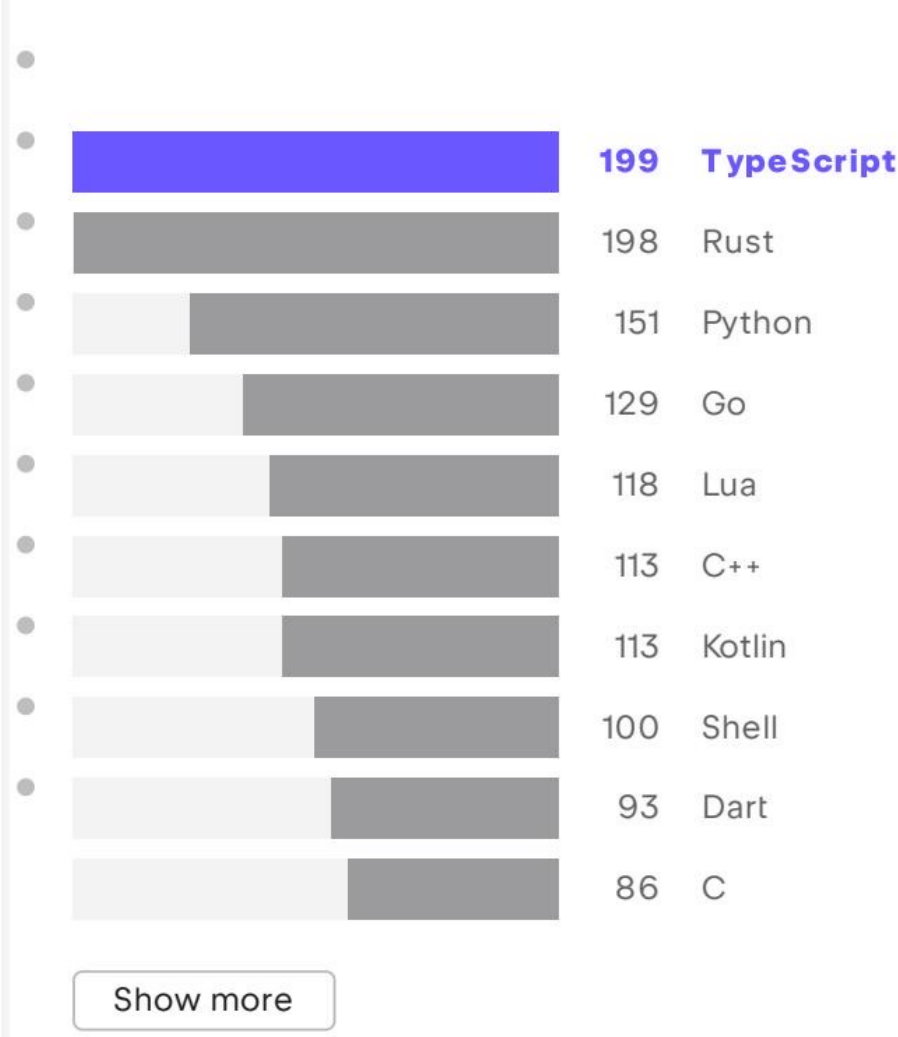
Combines memory safety with C++ level performance.
Cult-like following for systems programming.

3. Python (151 Score)

Loved for its simplicity and vast ecosystem.
The de-facto standard for AI/ML development.

The Long Tail

Go (129), Lua (118), C++ (113), Kotlin (113), Shell (100).



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven heavily by application domain, not just general preference or syntax.

Domain Dominance Breakdown

Python: Undisputed Data Leader

AI/ML (31%) and Data Science (28%).
The core engine of modern data analytics.

JS/TS: Web UI Monopoly

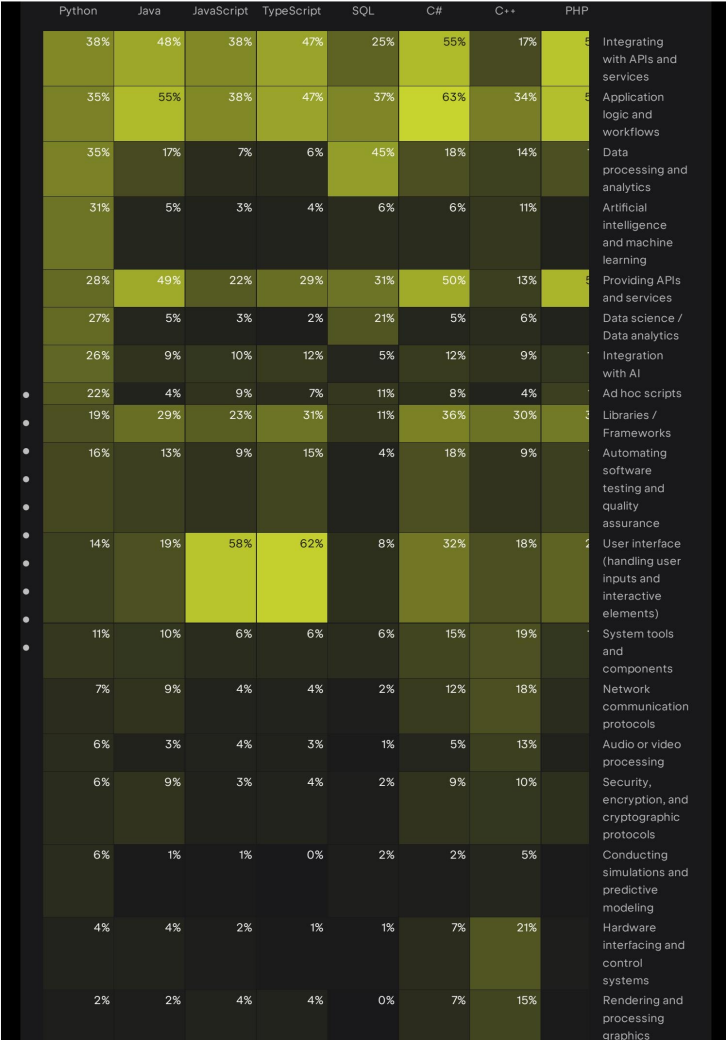
TypeScript (62%) and JavaScript (58%).
Effectively own all user interface development.

Java: Enterprise Backbone

APIs & Services (49%).
Maintains dominance in robust application logic.

C++: Low-Level Systems

System Tools & Components (27%).
Unmatched in hardware interfacing and graphics.



What This Means for Your Technology Strategy in 2025

Align hiring, retention, and strategic bets with structural shifts in language dominance and developer satisfaction.

1. Invest in Python for AI/ML

It is the undisputed leader for data science and AI/ML workloads. Align ML talent pipelines heavily with Python.

2. TypeScript is the Standard for Web UI

TypeScript represents the future of frontend and full-stack development, delivering necessary safety to the JS ecosystem.

3. Go and Rust for Infrastructure

These are the primary growth plays for backend, systems, and performance-critical infrastructure, commanding high developer loyalty.

4. Transition Away from PHP and Ruby

These hiring pools are shrinking rapidly. Future-proof new projects by selecting stacks with growing adoption and satisfaction.

Actionable Retention Strategy: Factor type safety and modern tooling into your tech choices, as they strongly correlate with developer satisfaction.